

C-UAS Training Aid Demo

Unmanned Aerial Detection System

1. Sensor Detection Subsystem

#	Component	Specification	Price (BDT)
1.1	mmWave Radar	3X TI IWR6843 / AWR1843 Radar Module	150,000
1.2	IR PTZ Camera	Night Vision Tracking Camera	50,000
1.3	Acoustic Detection Array	4 MEMS Microphone Array	50,000
1.4	LRF	2000m Laser Rangefinder	50,000
1.5	RF Detector Module	2400–2500 MHz	100,000

Detection Subtotal: 400,000

2. AI Processing & Control

#	Component	Specification	Price (BDT)
2.1	AI Computing Core	NVIDIA Jetson Nano 4GB Module	60,000
2.2	Microcontroller	STM32F4 Control Board	50,000
2.3	Storage & Interface	SSD, Communication Interface	10,000

Processing Subtotal: 120,000

3. Command & Monitoring Console

#	Component	Specification	Price (BDT)
3.1	Display Unit	10" Industrial Display	15000
3.2	Operator Keyboard	Rugged Keyboard	2000
3.3	Joystick Controller	PTZ Control Joystick	3000

Console Subtotal: 20,000

4. Power & Communications

#	Component	Specification	Price (BDT)
4.1	Power Supply	Custom PSU	10000
4.2	UPS Backup	Battery Backup System	10000
4.3	Communication Module	Ethernet / Wi-Fi	10000

Power & Communication Subtotal: 30,000

5. Mechanical Structure

#	Component	Specification	Price (BDT)
5.1	Weatherproof Enclosure	IP65 Rated	50000
5.2	Mechanical Frame	Tower & Mounting Structure	50000

Mechanical Subtotal: 100,000

6. Software Development & Integration

#	Component	Specification	Price (BDT)
6.1	AI Detection Software	YOLO-Based Drone Detection	30,000
6.2	Dashboard & Logging	Monitoring Interface	20,000
6.3	Integration & Testing	System Calibration	30,000

Software Subtotal: 80,000

Budget Summary

Category	Cost (BDT)
Sensor Detection Subsystem	400,000
AI Processing & Control	120,000
Command & Monitoring Console	20,000
Power & Communications	30,000
Mechanical Structure	100,000
Software Development & Integration	80,000

Total Project Budget 750,000

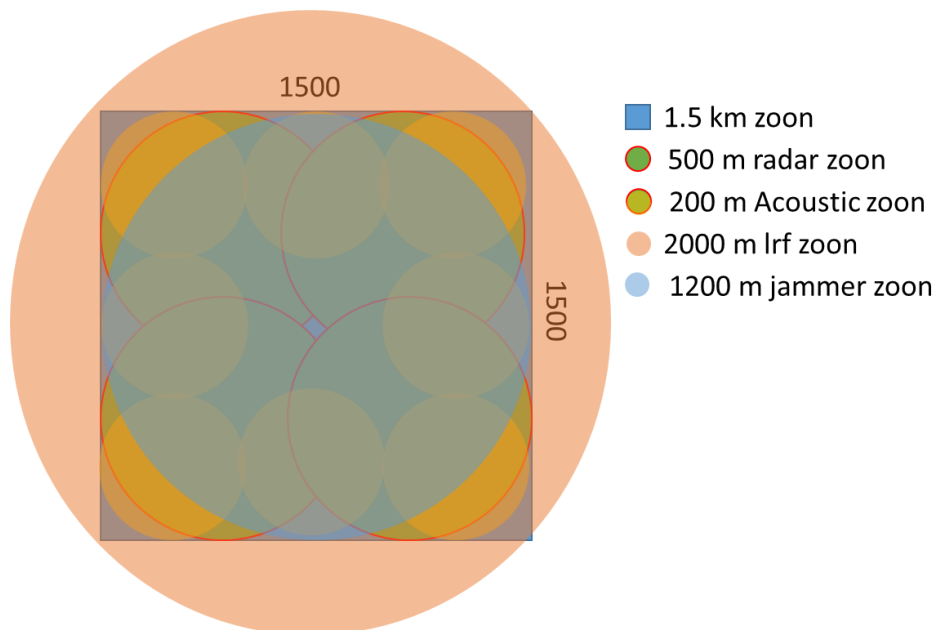
7. To further improve the system's operational effectiveness, we recommend the integration of the following optional subsystems.

#	Component	Price (BDT)
7.1	RF Power Amplifier (Jammer Module) per modet	100,000
7.2	1 Additional Radar Module	50,000
7.3	GNSS SPoofer Module	100,000

Optional Subsystems: 250,000

Technical Capability

Capability	Expected Performance
Radar Detection	1500 m
Visual Identification (Day)	1.2 m
Visual Identification (Night)	800 m
Acoustic Detection	2000 m
RF Detection Range	2km (passive)
Jamming Effective Range (2.4GHz)	1500 m
Laser Rangefinder	2000m
Jamming Effective Range (Future Multi-Band)	1500 m
Object Classification	High for visible targets
Tracking Accuracy	$\pm 10-15^\circ$
Multi-Target Tracking	20-30 Targets
Update Rate	10-30 Hz
Operating Hours	24/7
Power Requirement	12-24 VDC



Note: This system is a detection, tracking, and awareness trainer prototype. It is designed for research, testing, and educational purposes.

Technical Specification Section

mmWave Radar Subsystem

Parameter	Specification
Model	TI IWR6843 / AWR1843
Radar Type	FMCW mmWave Radar
Frequency Band	60 GHz (IWR6843) / 76–81 GHz (AWR1843)
Bandwidth	Up to 4 GHz
Source	https://mou.sr/4fcJNor
Range Resolution	4–15 cm
Velocity Measurement	Yes
Azimuth Measurement	Yes
Elevation Measurement	Optional
Update Rate	Up to 30 Hz
Detection Range	100–500 m
Multi-Target Tracking	Up to 20 Targets
Interface	UART / SPI / CAN / Ethernet
Operating Temperature	-40°C to +85°C
Power Consumption	<10 W

Embedded Controller

Parameter	Specification
MCU Family	STM32F4
CPU Frequency	216 MHz
Flash Memory	Up to 16 MB
SRAM	Up to 4 MB
Communication	UART, CAN, SPI, I2C, USB
Operating Voltage	3.3 V
Real-Time Processing	Yes
Source	Coustom

Electro-Optical / IR PTZ Camera

Parameter	Specification
Sensor Type	CMOS Sensor
Resolution	1920 × 1080
Frame Rate	30 FPS
Optical Zoom	20×
IR Illumination Range	100–200 m
Detection Range (Day)	Up to 800 m
Detection Range (Night)	Up to 300 m
Pan Rotation	360° Continuous
Tilt Rotation	±90°
Video Compression	H.264 / H.265
Interface	Ethernet
Power Consumption	<20 W
Source	https://luminixsky.com/product/skydroid-c20-three-axis-night-vision-gimbal/

Acoustic Detection Array

Parameter	Specification
Sensor Type	MEMS Microphone Array
Number of Microphones	8 Elements
Frequency Response	20 Hz – 20 kHz
Acoustic Detection Range	100–200 m
Direction Finding	Yes
Sampling Rate	48 kHz
Interface	USB / I2S
Power Consumption	15 W
Source	Coustom

AI Computing Module

Parameter	Specification
Module	NVIDIA Jetson Orin 4GB
AI Performance	Up to 20 TOPS
CPU	6-Core ARM Cortex-A78AE
GPU	NVIDIA Ampere Architecture
RAM	4 GB LPDDR5
Storage	128 GB SSD
Operating System	Ubuntu Linux
AI Frameworks	TensorFlow, PyTorch, YOLOv8
Power Consumption	7–15 W
Source	https://store.roboticsbd.com/development-board/1373-nvidia-jetson-nano-developer-kit-b01-4gb-robotics-bangladesh.html

Command & Control Console

Parameter	Specification
Display	15" LCD Touch Screen
Resolution	1280 × 800
Control Device	TPC156T0001G01V1
Keyboard	Industrial USB
Video Channels	4 Simultaneous
Data Logging	Yes
Alarm Display	Yes
Source	https://www.dwin-global.com/15-6-inch-g-g-structure-tempered-glass-capacitive-touch-screen-tpc156t0001g01v1-product/

Communication System

Parameter	Specification
Ethernet	10/100/1000 Mbps
Wi-Fi	IEEE 802.11 b/g/n/ac
GPS	Multi-GNSS
Protocols	TCP/IP, UDP, MQTT
Remote Monitoring	Supported
Source	Coustom

Power System

Parameter	Specification
Input Voltage	220 VAC
Output Voltage	12 VDC / 24 VDC
Battery Backup	2–4 Hours
UPS Type	Online / Line Interactive
Surge Protection	Yes
Power Consumption	80–150 W
Source	Coustom

Overall System Performance

Capability	Performance
Drone Detection	100–500 m
Drone Classification	85–95%
Multi-Target Tracking	5–20 Targets
Detection Latency	<3 sec
Tracking Accuracy	±10–15°
Continuous Operation	24/7
Weather Resistance	IP65
System Availability	>95%